Asmit Bhattacharyya

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Portfolio | LinkedIn | GitHub

Professional Summary

Aspiring AI/ML Engineer with hands-on experience developing YOLO-based object detection and tracking pipelines, LLM-powered mathematical reasoning systems, and AI-driven workflow automation. Proficient in Python, PyTorch, OpenCV, FastAPI, and LLM APIs, with experience integrating symbolic computation using SymPy for solution verification, building multi-provider LLM architectures, and developing JavaScript-based API integrations across enterprise platforms.

Technical Skills

Programming Languages: Python, C++, JavaScript
Machine Learning & Computer Vision: Machine Learning, Deep Learning, Computer Vision, Object Detection, Object Tracking
AI/ML Libraries & Frameworks: PyTorch, YOLO, Ultralytics, OpenCV, Roboflow, Supervision, SymPy, NumPy
LLM & AI Application Development: LLM APIs, OpenAI Integration, Prompt Engineering
Backend & APIs: FastAPI, Django, Node.js, REST APIs
Databases: PostgreSQL, MySQL, Redis
Frontend: Next.js, Vue.js, HTML, CSS

Professional Experience

AppseAI | Kolkata, India — March 2026 – August 2026
Techno-Functional Apprentice

- Designed and developed aproximately 100 integration workflows connecting enterprise systems and third-party applications, including SAP, Microsoft Dynamics 365 Business Central, HubSpot, and Shopify.
- Implemented custom JavaScript for data transformation, business rules, API interactions, and workflow automation across connected business systems.
- Integrated OpenAI-powered capabilities into automated workflows to support AI-enabled business processes.
- Tested, troubleshot, and debugged integrations, API interactions, and data synchronization across connected business systems.

Education

MCKV Institute of Engineering (affiliated to MAKAUT) — Sep 2022 – June 2026
B.Tech. in Computer Science & Engineering – AI&ML

Projects

Football Match Analysis
Python, YOLO, Ultralytics, OpenCV, Roboflow, Supervision, Streamlit

- Developed a computer vision pipeline using YOLO to detect and track football players, referees, and the ball across match footage.
- Implemented team assignment, ball tracking, player positioning, and tactical-map visualization to derive football analytics from video data.
- Extended the analysis pipeline into a Streamlit web application, providing automated video analysis and visual representation of match events and player positions.

MathForge
Python, FastAPI, SymPy, NumPy, LLM APIs, Astral UV

- Developed an AI-powered application for generating and solving mathematical questions across configurable topics, difficulty levels, and answer formats.
- Integrated LLM-generated solutions with symbolic computation using SymPy to support mathematical answer verification.
- Implemented a multi-provider LLM architecture with configurable models and generation parameters.